Diabetes Prediction Model using Machine Learning

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Abstract

Chronic metabolic disorder diabetes affects worldwide populations with high levels of blood sugar and serious health threats like heart disease and stroke. Rapid screening and response are important to alleviate the future complications linked to diabetes; still many available methods are pricey and inaccessible to individuals residing in remote locations. Machine learning (ML) approaches were applied in this research to build efficient models for detecting diabetes at an early time. Multiple algorithms such as Random Forest and Decision Tree were used in the study. To cope with imbalanced data the models utilized several diabetes datasets alongside preprocessing techniques for feature selection and random oversampling. For some datasets the accuracy rates reached 100% when using Random Forest and Extreme Gradient Boosting. The results indicate that diabetes prediction accuracy can be greatly improved using ML methods.

Keywords: In the field of healthcare classification algorithms help make early diagnoses more precise.

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Introduction

A chronic disorder of metabolism called diabetes mellitus presents with elevated blood sugar that may result in life-threatening complications like nerve damage and eye loss. Treatment and prevention of this condition trigger major health challenges that involve a wide range of populations. Based on data from the World Health Organization diabetes affected 108 million people in 1980 and by 2014 had surged to 422 million with the majority of cases taking place in countries with lesser income. By 2045 the International Diabetes Federation expects this number could grow to 700 million. Diabetes now ranks among the top killers worldwide and accounts for about 1.6 million annual fatalities. Urging to treat diabetes involves heavy expenses that reach nearly $760 billion around the globe in 2019.

Diabetes is primarily categorized into three types: There are types 1 and 2 diabetes and gestational diabetes. In Type 1 diabetes the immune system destroys the insulin-producing beta cells in the pancreas which causes low levels of insulin. The leading form of diabetes is Type 2 diabetes which develops when the body is unable to produce enough insulin or become insensitive to it. During pregnancy gestational diabetes occurs and boosts chances of developing Type 2 diabetes later on.

As diabetes becomes more common and is accompanied by high treatment expenditures there is great urgency for early detection and effective interventions. Classic diagnostic approaches such as the OGTT and HbA1c rely on complex instruments and time which might be unavailable in distant locations. To minimize the worldwide effects of this illness effectively requires the advancement of economical and accurate testing methods.

Over the past few years machine learning has become an effective instrument for disease prediction and diagnosis including diabetes. In machine learning frameworks large swaths of health data are examined for the discovery of trends and the prediction of outcomes. Different research projects examined how machine learning methods like Artificial Neural Networks and Random Forest aid in forecasting diabetes based on characteristics such as blood pressure and glucose levels. The models provide encouraging levels of accuracy as shown by various studies exceeding 90% in diabetes prediction.

Dealing with large datasets is a major problem in machine learning that may cause overfitting and raise the complexity of calculations. To tackle this challenge were applied feature selection methods that eliminate extra and unimportant features to boost the effectiveness of the predictive algorithms. This research seeks to use machine learning techniques and feature reduction to create a budget-friendly and user-friendly model for forecasting diabetes in its initial phases.

Literature Review

Diabetes is acknowledged to be one of the biggest health concerns that modern India faces, with millions of people suffering from Type II diabetes. India specifically has a high prevalence of diabetes, which scientists believe might be partly due to genetic factors: today, India has a young population where a high proportion is urbanized, consuming more processed foods than in the past. Recently, the studies on diabetic risks and their early identification and control in the Indian population have gained high research relevance; the increasing role of ML approaches and Indian datasets for this population is also noted.

Existing Traditional Epidemiological Studies and Unconventional Techniques in India
Indian researchers have carried out a lot of research on epidemiology of diabetes among the urban and rural inhabitants. Using data from the INDIAB study conducted in 2011, the Indian Council of Medical Research (ICMR) pointed out data regarding regional variation of diabetes and prediabetes in India. Risk factors which postulated as major causative agents of diabetes were identified as including diet, physical inactivity and stress the study asserted that the diabetes burden in urban regions related especially to lifestyle diseases. Historical tools of diagnosing diabetes include the OGTT and HbA1c but these have been augmented with new tools for diagnosing every diabetic patient especially in a high risk population.

Studies on the Risk Determinants of Diabetes in India
A plethora of research carried out in a few years from India has also recognized and proved various risk factors associated with the Indian population only. For example, Ramachandran et al., 2004 pointed out that Indians commonly develop diabetes at a lower BMI as compared to western counterparts attributable to high rates of central obesity, diabetes resistance consistently reported among this population. It has been evidenced from various studies that family history, hypertension and raised triglyceride levels play important role for diagnosing diabetes in Indians.Such studies came from professionals and institutions like the Madras Diabetes Research Foundation (MDRF) and revealed newer factors such as the waist-to-hip ratio and abdominal obesity as better predictors in Indians than global risks for the said disease. It has led to developments of more specific risk models with a view of comprehensively touching on the metabolic nature of the Indians.

Applications of Machine Learning in India
Recently, growing availability and utilization of healthcare data has motivated Indian researchers to build machine learning models to enhance the accuracy of diabetic predictions. Random Forest and Logistic Regression models were used by Mohan et al.

(2019) to learn about diabetes prediction from a dataset that included other cultural risk determinants such as diet, physical activity, and genetics. Collectively, the researchers realized that models that were trained using localized datasets went a long way in outperforming those based on global data and long listed the necessity for India specific models.

Using PimaIndian Diabetes Dataset as baseline and adding extra features which are frequently encountered in Indian clinical practice such as level of HbA1c, BMI and history of having diabetic family members, Saxena et al (2020) also designed a model. Their study showed that the integration of decision trees and boosting methods performed a high accuracy, useful for the screening of both the large-pocketed city dwelling and rural dwellers.

Use of Genetic Markers in Indian Research

The 8-10 genetic markers used in Indian research are located on chromosomes 2, 4, 6-7, and X. Indian researchers only started looking into the use of genetic markers for diabetes in the recent past. Scientific research done by Anjana et al. (2015) has described genes for TCF7L2 and KCNJ11 that are strong risk factors for Type 2 diabetes in Indians. These studies emphasize that although age, obesity and other conventional risk factors are imperative, there is a strong hereditary influence of having diabetes. However, the present scenario fails to offer routine genetic screening throughout the nation primarily owing to costs and structural constraints.

Wearable technology and Telemedicine in Indian Research

Telmedicine and wearable devices have now started getting importance in the Indian context as tools to monitor and predict diabetes.. Kapur et al. (2021) examined a feasibility Randomised Controlled Trial on the effectiveness of CGMs combined with mHealth platforms to forecast and control diabetes in real time terms. It also discovered that the use of wearables alongside ML-based prescriptive models assisted in the stabilization of blood sugar levels and issued early alerts of diabetic complications leading to decreased hospitalization.

Telemedicine platforms have also been integrated across various studies in India especially during the COVID-19 pandemic. A trial study conducted by Srivastava et al., (2020) integrated telehealth services where the user's daily glucose record and their lifestyle data were utilized to forecast the user's likelihood of having diabetes . Therefore, the research focused on adopting the machine learning algorithm in conjunction with ongoing data monitoring in wearable platforms, and suggested the findings could go a long way in early diagnosis of diabetes in rural areas, where access to relevant health facilities is scarce.

This paper also presents possible challenges and limitations in Indian studies.
Despite the advances that have been made on prediction of diabetes using machine learning in India several challenges still persist. One challenge is the inability of researchers to access clean and comprehensive data that will capture the diverse population of India. The work done in several centers mainly target urban dwellers restricting the findings to other populace especially those in rural settings. However, the samples used in many of these research also have small size and therefore the researcher can only generalize the conclusion with caution. Two of the remaining limitations are data privacy and the cost of deploying such machine learning models in a public health environment.

Existing Method

Existing System

1. BeatO
BeatO enables Indian users to monitor their blood sugar with a smartphone peripheral that interfaces with a glucometer. It delivers specific suggestions and dietary choices. This app integrates the latest glucose information and provides suggestions for adjusting daily habits to relieve the effects of diabetes. A platform for health insurance and options for scheduling doctor visits and controlling prescriptions are both present.

2. Diabeto
The mobile platform named Diabeto was created for managing diabetes. Users access their glucose readings by connecting toBluetooth devices for different glucometers. The app delivers analytics and visualizations made from the saved data. The goal of Diabeto is to enhance user comprehension of their state and to give healthcare workers online access to their patients' information for improved control.

3. HealthifyMe
With a focus on health and exercise management for diabetes forecasting and assistance purposes The app provides tailored eating and exercise schedules and supports users in tracking calories with diabetes or concerning health. AI generates its recommendations while health specialists and nutritionists provide consultations. Continuous oversight for those managing chronic conditions like diabetes is available in HealthifyMe's premium service.

4. Wellthy Care
Utilizing AI technology and focused on managing diabetes and other chronic conditions is Wellthy Care. The platform delivers constant results and individual guidance rooted in collected health data. The application delivers daily suggestions and tracks user progress along with alerts for medication intake.

5. PhableCare
As an Indian application PhableCare enables users to handle chronic disorders such as diabetes by engaging with doctors and recording their health indicators while prompting users to take their prescribed medication. The app connects with wearable gadgets and glucose meters to gather information on blood sugar and vital signs while offering an all-round health perspective. Through its platform Phable lets users

connect with healthcare professionals via virtual conversations and delivers health information influenced by AI along with personalized care recommendations.

6. DiabetesMitra
The app aims to support efforts in diabetes treatment and avoidance. It enables users to watch their blood sugar and obtain dietary suggestions for better physical activity.The tool furnishes instructional aids that aid in comprehending and overcoming health concerns. In DiabetesMitra. Health data undergoes processing by AI and machine learning for delivering individual wellness advice.

7. SugarFit
Individual programs that aim to improve Type 2 diabetes are designed by the diabetes management platform of SugarFit. Through the app users can monitor their glucose levels and receive tailored guidance along with keeping health records. Using information from users to tailor nutrition suggestions and exercise routines the app assists individuals in lowering their medication dependence and managing their health better.

Motivation

The need to develop this type of diabetes prediction model is rooted in the fact that millions of residents in India are at risk of developing the condition on account of genetic factors, lack of exercise, and unhealthy diet. Primary care is an important approach to the prevention of chronic health hazards, but current systems of health care do not accurately identify targeted individuals until clinical symptoms have manifested themselves.

Rising Prevalence of Diabetes: India is experiencing a fast moving phase of diabetes epidemic. This model is intended to assist in early identification of disease states and inhibit the development of further associated issues and worse control of DM.

Late Diagnosis and Its Consequences: Far too many are diagnosed too late that invariably result in extreme complications. The model provides risk identification at the initial stage to facilitate appropriate intervention.

Personalized Risk Prediction: It offers personalized risk centered on specific features such as BMI and HbA1c to adapt to the distribution of risk variables in the India population.

Accessibility of Healthcare: Lack of access to healthcare is the major problem in rural areas. They are comfortable with using text messages and the model can give them an accurate risk prediction at a very low cost.

Reducing Healthcare Burden: By using the kind, early diagnosis, the Indian health care system can be made less pressured by early and insignificant early health complications that require intensive mechanisms of treatment.

Contribution to Healthcare: The model enhances preventive care because it fosters early detection, efficient distribution and use of the resources required, and easier hospital loads.

Challenges to Healthcare Infrastructure: It offers a kind of low-tech solution to geographical locations where either docs, nurses, hospitals, and other healthcare facilities are scarce.

Cost-Effective Solution: The model is easier and cheaper than expensive diagnosis tools, incorporating easily available health identifiers such as BMI and glucose levels.

Preventive Health Focus: Moreover, by detecting such patients ahead of time the model encourages a change in behavior and seeking of medical help which would reduce future overall healthcare expenses.

Proposed Method With Architecture

The present technique seeks to diagnose diabetes from the patient data by applying the machine learning techniques. These include after data collected and cleaned, used methods like Histograms and Heat map to analyze the relationship of several features. Gaussian Naive Bayes, KNN, SVM, Decision Tree, Random Forest and Logistic Regression were used as the algorithms for training the model. More advanced ensemble methods such as voting and stacking were also used in an attempt to increase reliability. The performance of each model was measured applying indicators such as accuracy and F1-score; the best algorithm was identified to make accurate diabetes predictions.

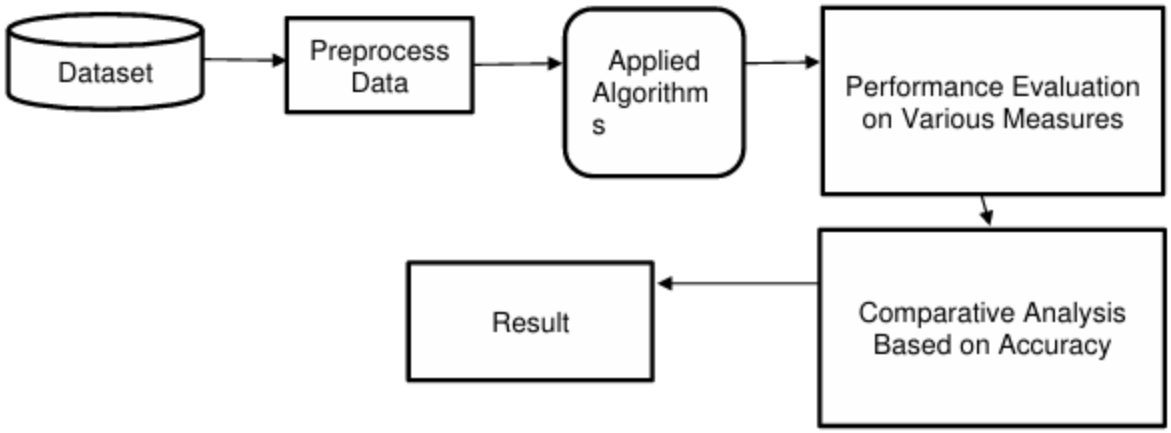


Fig - Proposed Model Diagram

Work Flow of Model-

1. **Data Collection:**
Collect patient health information such as the BMI, Level of Hypertension and the HbA1c feature which is important for diabetes diagnoses.

2. Data Preprocessing:
Pre-process the data means to make the necessary preparation for making the data ready for feeding into Machine Learning algorithms. It means that in this step we have to handle missing values and normalize features.

3. Data Visualization:
All of the above, histograms, heatmaps, density plots, box plots and pair plots should be used to assess the distribution and correlation of features. These insights inform the construction of the model.

4. Algorithm Implementation:
Train several machine learning algorithms: NB Gaussian, KNN, SVM, Decision tree, Random forest and logistic regression. These algorithms ALL give a different solution to the classification problem.

5. Ensemble Techniques:
Vote and stack the ensemble methods in order to increase accuracy by using the result of other models in a project. This assists in strong points of various algorithms.

6. Model Evaluation:
Assess the performance of each model using classification of model's strengths, weakness on accuracy, precision, recall, and F1-score for test on diabetes.

7. Best Model Selection:
Using the metrics in the above section we choose the best algorithm or an ensemble method for accurate prediction of diabetes.

Methodology

DataSet Description

The diabetes dataset originated from a website called "Kaggle". The Diabetes dataset contains a total of 100000 cases. The objective is to predict based on the measures to predict if he or she is diabetic or not.

It contain features like

Gender: In essence, studies suggest that gender plays a role in diabetes in the following ways: men have a higher rate of Type 2 diabetes than women; this is probably attributable to a difference in body and fat distribution. Awareness of gender means that risk evaluations can be applied more correctly based on gender relations.

Age: They determined that age is a risk factor that has been confirmed times without number to be directly proportional to the risk of diabetes. And as expected the older persons are more likely at higher risk due to the compounded effects of their lifestyle and altered physiological functions that contribute to alteration of insulin function and glucose tolerance.

Hypertension: High blood pressure often coexists with metabolic syndrome, which refers to factors that raise the risk of diabetes. People with hypertension have higher tendencies of having insulin resistance which is why it's an important parameter to consider when predicting diabetes.

Smoking History: We have seen that smoking increases the chances of getting Type 2 diabetes. Some of the chemicals in cigarettes can cause a condition known as insulin resistance and high blood sugar levels. Proper smoker status is placed in the model to help learn the lifestyle that leads to the set disease.

Heart Disease: Diabetes can be a danger for people who already had a heart disease because they have a greater risk of getting it. Heart disease and diabetes can cause each other since one can worsen the other. In this feature, there is an appreciation of general cardiovascular health and how it affects diabetes risk.

BMI (Body Mass Index): As we know, BMI defines fat and obesity that are important factors concerning diabetes. The results also show the mean BMI values are elevated and are related to insulin resistance. By monitoring BMI, the model is able to determine the weight-associated dangers proficiently.

HbA1c Level: HbA1c also known as glycated hemoglobin measures blood glucose control over the last 2-3 months. High HbA1c level is an index of suboptimal glycemic regulation and is employed in the diagnosis of diabetes. This feature is very important in determining long-term glycemic control.

Blood Glucose Level: Random blood glucose measurements are important for dividing people into those requiring urgent diabetes indications. The data shows that greater values may reflect on insulin insensitivity and risk of developing Type 2 diabetes. The information offered through this feature helps to assess an individual's live metabolic rates.

	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
0	Female	80.0	0	1	never	25.19	6.6	140	0
1	Female	54.0	0	0	No Info	27.32	6.6	80	0
2	Male	28.0	0	0	never	27.32	5.7	158	0
3	Female	36.0	0	0	current	23.45	5.0	155	0
4	Male	76.0	1	1	current	20.14	4.8	155	0

Fig - Dataset

Data Processing and Visualization

Data Processing

Data preprocessing is the process of preparing original data for ML. It is indeed the most important step in the process of developing an ML model. It is a necessary step for ML algorithms to improve the model's accuracy and efficiency. Data preprocessing covers data preparation, which includes data integration, cleansing, normalization, and transformation, as well as data reduction activities such as feature selection, instance selection, and discretization. We have conducted some data analysis after collecting the data. Then, we have done preprocessing tasks including outliers removal and dealing with missing values, data normalization, encoding, and so on.

1.Outliers handling: A dataset may contain extreme values that are beyond the acceptable limits and dissimilar to the rest of the data. This kind of data may reduce the performance of the ML algorithm. Any value ≤ Q1 −1.5×interquartile range (IQR) or ≥ Q3 − 1.5 × IQR is considered an outlier.

2.Missing value handling: One of the most difficult tasks for analysts is dealing with missing values, because making the proper decision on how to deal with them leads to more robust data models. We can handle missing values in various ways, such as ignoring the 6 DIGITAL HEALTH row, imputation of the missing values with data means, median or mode of the observation.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 9 columns):
 #   Column               Non-Null Count   Dtype
---  ------               --------------   -----
 0   gender               100000 non-null  object
 1   age                  100000 non-null  float64
 2   hypertension         100000 non-null  int64
 3   heart_disease        100000 non-null  int64
 4   smoking_history      100000 non-null  object
 5   bmi                  100000 non-null  float64
 6   HbA1c_level          100000 non-null  float64
 7   blood_glucose_level  100000 non-null  int64
 8   diabetes             100000 non-null  int64
dtypes: float64(3), int64(4), object(2)
memory usage: 6.9+ MB
```

Fig - Information of Dataset

	age	hypertension	heart_disease	bmi	HbA1c_level	blood_glucose_level	diabetes
count	100000.000000	100000.00000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000
mean	41.885856	0.07485	0.039420	27.320767	5.527507	138.058060	0.085000
std	22.516840	0.26315	0.194593	6.636783	1.070672	40.708136	0.278883
min	0.080000	0.00000	0.000000	10.010000	3.500000	80.000000	0.000000
25%	24.000000	0.00000	0.000000	23.630000	4.800000	100.000000	0.000000
50%	43.000000	0.00000	0.000000	27.320000	5.800000	140.000000	0.000000
75%	60.000000	0.00000	0.000000	29.580000	6.200000	159.000000	0.000000
max	80.000000	1.00000	1.000000	95.690000	9.000000	300.000000	1.000000

Fig - Description of Dataset

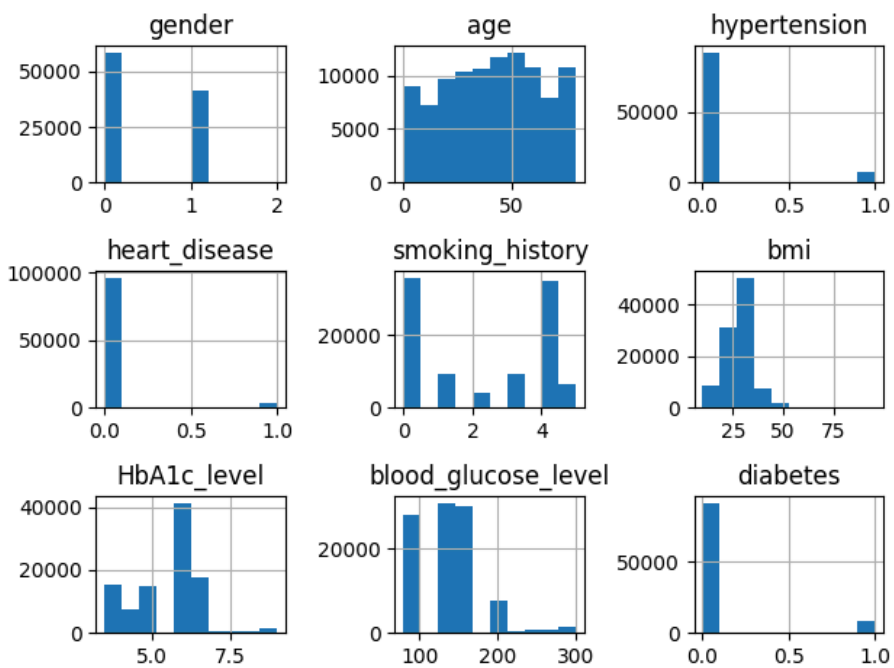
3.Label encoding: All the data entered in the ML algorithm must be numerical. However, in most cases, the dataset may contain categorical data rather than numerical values. So, if the dataset contains any categorical attribute, it must be needed to convert to numeric values before fitting and evaluating an ML model. Label encoding is the process of transforming labels of text/categorical values into a numerical format which is understandable by the ML algorithms.

4.Standardization: When features of an input dataset have considerable variations between their ranges or when they are collected or measured in different measurement units, standardization becomes necessary. Differences hamper the performance results for ML models in the range of initial features. So normalization or standardization can solve this issue and improve the prediction quality.

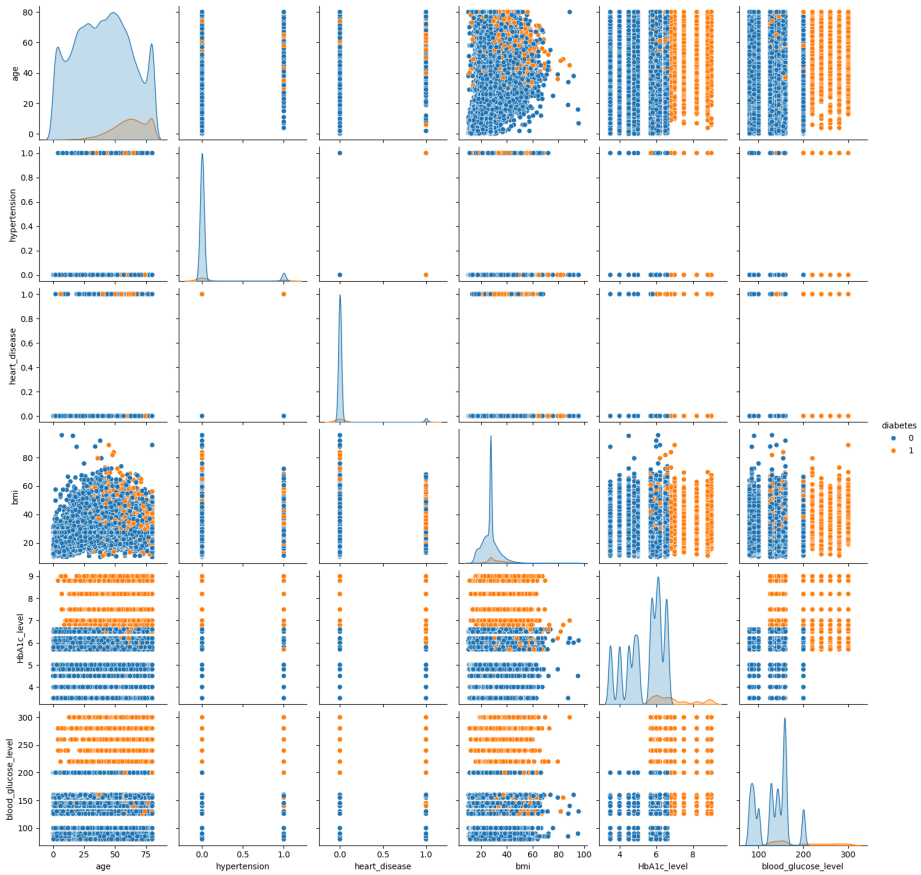
Data Visualization

Data visualization is the process of using figures to understand the collected data. In creating the ML model, this is one of the processes it allows to determine patterns, trends, and correlation in data. It also allows visualizing the data that underlies the development of our model and our decisions.

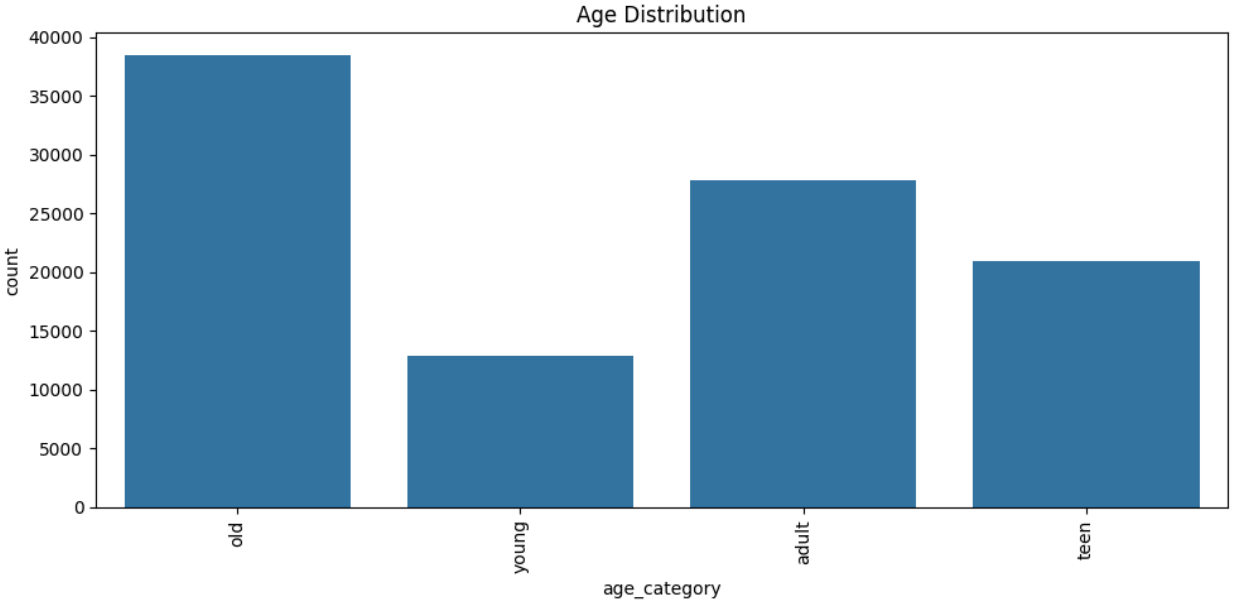
1.Histogram



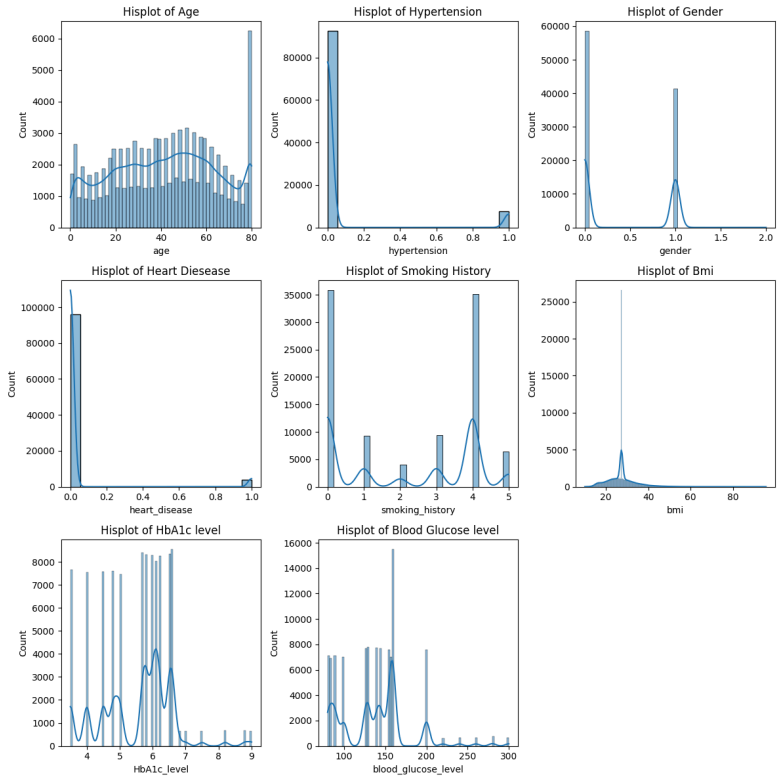
2.Pairplot



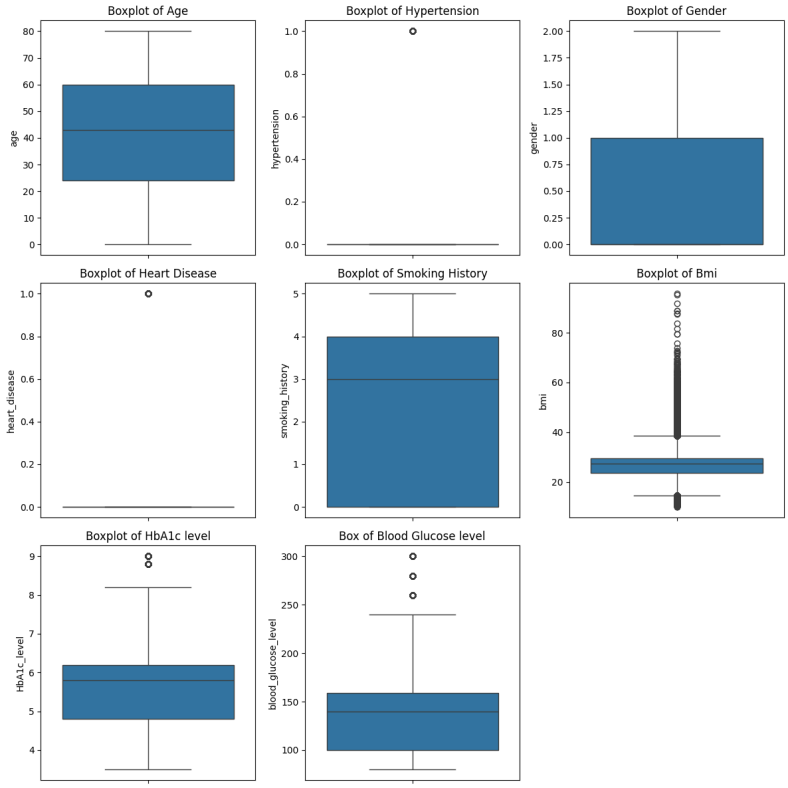
3.Age Distribution Graph



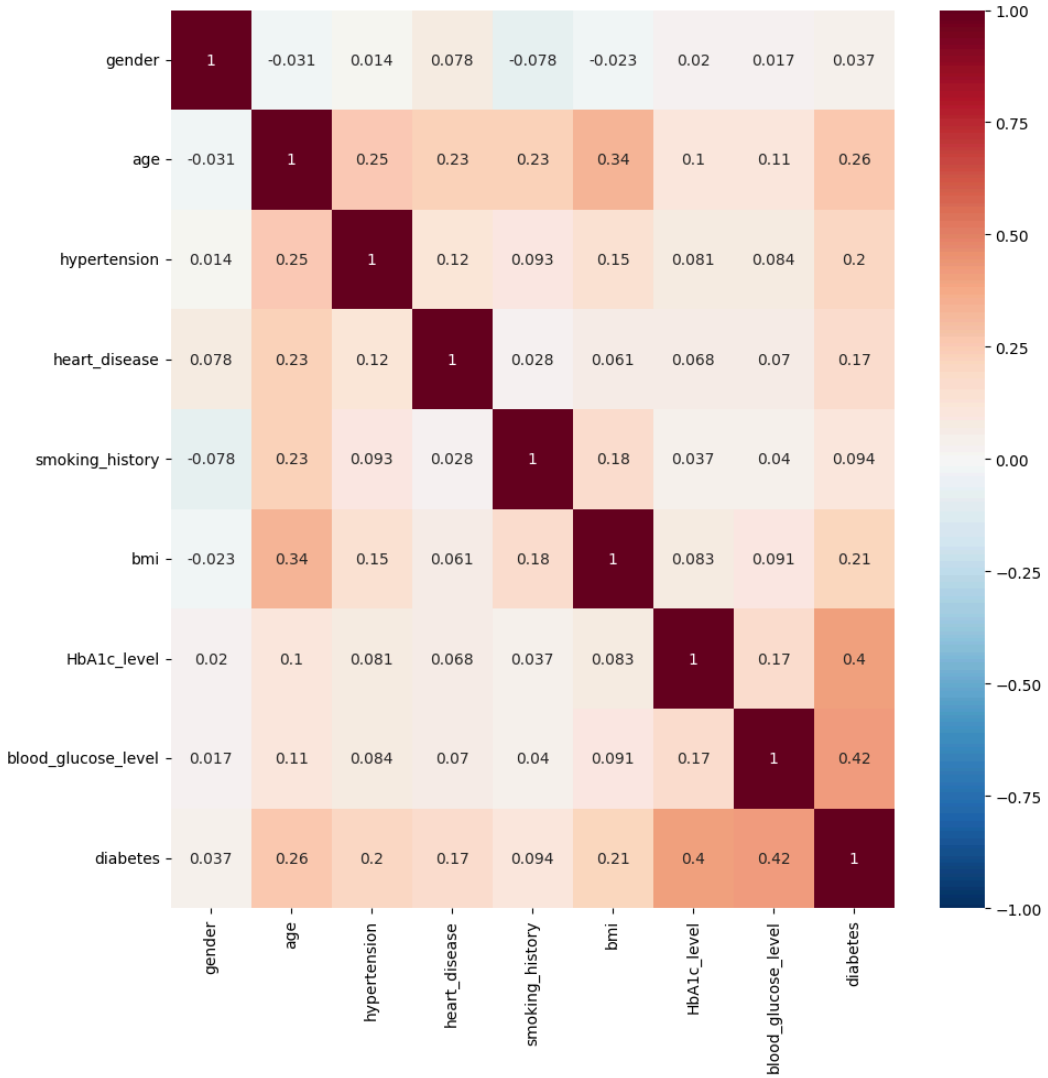
4.Histplot



5.Boxplot



6.HeatMap


gender
age
hypertension
heart_disease
smoking_history
bmi
HbA1c_level
blood_glucose_level
diabetes
1 -0.031 0.014 0.078 -0.078 -0.023 0.02 0.017 0.037
-0.031 1 0.25 0.23 0.23 0.34 0.1 0.11 0.26
0.014 0.25 1 0.12 0.093 0.15 0.081 0.084 0.2
0.078 0.23 0.12 1 0.028 0.061 0.068 0.07 0.17
-0.078 0.23 0.093 0.028 1 0.18 0.037 0.04 0.094
-0.023 0.34 0.15 0.061 0.18 1 0.083 0.091 0.21
0.02 0.1 0.081 0.068 0.037 0.083 1 0.17 0.4
0.017 0.11 0.084 0.07 0.04 0.091 0.17 1 0.42
0.037 0.26 0.2 0.17 0.094 0.21 0.4 0.42 1
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Algorithm Selection

As now we have successfully completed Data processing and data visualization, now it's time to find the best ML algorithm which provides us maximum accuracy. And will be suitable for our model.

We have used seven different ML classifiers to train the model, including DT, K-nearest neighbor (KNN), LR and SVM, Random Forest and predict diabetes. Depending on several performance metrics, the performance of each classifier has been analyzed. The following subsection will describe the ML algorithms used for the predictive model.

Support Vector Machines (SVM)

Support Vector Machine (SVM) is a powerful machine learning algorithm used for linear or nonlinear classification, regression, and even outlier detection tasks. SVMs can be used for a variety of tasks, such as text classification, image classification, spam detection, handwriting identification, gene expression analysis, face detection, and anomaly detection. SVMs are adaptable and efficient in a variety of applications because they can manage high-dimensional data and nonlinear relationships.

k-Nearest Neighbors (k-NN)

The K-Nearest Neighbors (KNN) algorithm is a supervised machine learning method employed to tackle classification and regression problems. Evelyn Fix and Joseph Hodges developed this algorithm in 1951, which was subsequently expanded by Thomas Cover. The article explores the fundamentals, workings, and implementation of the KNN algorithm.

Decision Trees

A decision tree is a type of supervised learning algorithm that is commonly used in machine learning to model and predict outcomes based on input data. It is a tree-like structure where each internal node tests on attribute, each branch corresponds to attribute value and each leaf node represents the final decision or prediction. The decision tree algorithm falls under the category of supervised learning. They can be used to solve both regression and classification problems.

Naive Bayes Classifiers

Naive Bayes classifiers are a collection of classification algorithms based on Bayes' Theorem. It is not a single algorithm but a family of algorithms where all of them share a common principle, i.e. every pair of features being classified is independent of each other

Random Forest

Random Forest algorithm is a powerful tree learning technique in Machine Learning. It works by creating a number of Decision Trees during the training phase. Each tree is constructed using a random subset of the data set to measure a random subset of features in each partition. This randomness introduces variability among individual trees, reducing the risk of overfitting and improving overall prediction performance.

Logistic Regression

Logistic regression is a supervised machine learning algorithm used for classification tasks where the goal is to predict the probability that an instance belongs to a given class or not. Logistic regression is a statistical algorithm which analyze the relationship between two data factors.

Ensemble Learning

Ensemble learning is a powerful technique in machine learning where multiple models are combined to improve overall predictive performance and robustness. The fundamental idea is that by aggregating the predictions of diverse models, the weaknesses of individual models can be mitigated, leading to more accurate and reliable results.

1.Stacking

Stacking is a way to ensemble multiple classifications or regression models. There are many ways to ensemble models, the widely known models are Bagging or Boosting. Bagging allows multiple similar models with high variance to be averaged to decrease variance. Boosting builds multiple incremental models to decrease the bias, while keeping variance small.

2.Voting

Voting in sci-kit-learn (Sklearn) allows us to combine multiple machine-learning modules and use a majority vote or a weighted vote to make predictions. It's a way to ensemble different models for potentially better performance.The Voting in Sklearn is an ensemble method that combines multiple individual classifiers or regressors to make predictions. It supports both hard and soft voting strategies. Now let's understand what ensemble learning is.

Model Selection

The prediction of any ML algorithm could have four distinct results depending on the confusion matrix as indicated : true positive (TP), true negative (TN), false positive (FP), and false negative (FN). Then, we consider the following metrics to analyze the proposal

Predicted results	Actual positive	Actual negative
Yes	TP	FP
No	FN	TN

Confusion Matrix

A confusion matrix is a matrix that summarizes the performance of a machine learning model on a set of test data. It is a means of displaying the number of accurate and inaccurate instances based on the model's predictions. It is often used to measure the performance of classification models, which aim to predict a categorical label for each input instance.

Accuracy

Accuracy measures the overall correctness of the model's predictions. It is calculated as the ratio of the number of correct predictions (TPs and TNs) to the total number of predictions made

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}$$

Precision

Precision quantifies the ability of the model to avoid FPs. It is calculated as the ratio of TPs to the sum of TPs and FPs

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall

Recall measures the model's ability to capture all positive instances. It is calculated as the ratio of TPs to the sum of TPs and FNs

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1 Score

TheF1-score is the harmonic mean of precision and recall, providing a balance between the two metrics. It is calculated as twice the product of precision and recall divided by the sum of precision and recall

$$\text{F1_score} = 2 * \frac{(\text{precision} * recall)}{(\text{precision} + recall)}$$

Classification Report

A classification report is a text summary that shows the main metrics for each class of a machine learning model. It usually includes the precision, recall, F1-score, and support for each class, as well as the weighted average of these metrics across all classes.

Algorithm Analysis

Algorithm Name	Test Accuracy
Naive Bayes Classifiers	0.9038666666666667
k-Nearest Neighbors (k-NN)	0.9622666666666667
Logistic Regression	0.9613333333333334
Decision Trees	0.9726666666666667
Random Forest	0.9726666666666667
Support Vector Machines (SVM)	0.9612666666666667

Conclusion

On the above Algorithm analysis we can say that Decision Trees and Random Forest give us the highest Accuracy among all Algorithms.

Ensemble Technique for Optimization

Ensemble learning is a machine learning technique that combines the predictions from multiple individual models to obtain a better predictive performance than any single model. The basic idea behind ensemble learning is to leverage the wisdom of the crowd by aggregating the predictions of multiple models, each of which may have its own strengths and weaknesses. This can lead to improved performance and generalization.

Uses of Ensemble Learning

Classification and Regression: Ensemble techniques make problems like classification and regression versatile in various domains, including finance, healthcare, marketing, and more.

Anomaly Detection: Ensembles can be used to detect anomalies in datasets by combining multiple anomaly detection algorithms, thus making it more robust.

Medical Diagnostics: In healthcare, ensembles can be used to make more accurate predictions of diseases based on various medical data sources and diagnostic models.

Climate Prediction: Ensembles of climate models help in making more accurate and reliable predictions for weather forecasting, climate change projections, and related environmental studies.

There are two Approaches of ensemble learning we have used in our model to enhances its accuracy

1.Stacking

Stacking is a way to ensemble multiple classifications or regression models. There are many ways to ensemble models, the widely known models are Bagging or Boosting. Bagging allows multiple similar models with high variance to be averaged to decrease variance. Boosting builds multiple incremental models to decrease the bias, while keeping variance small.

The point of stacking is to explore a space of different models for the same problem. The idea is that you can attack a learning problem with different types of models which are capable of learning some part of the problem, but not the whole space of the problem. So, you can build multiple different learners and you use them to build an intermediate prediction, one prediction for each learned model. Then you add a new model which learns from the intermediate predictions the same target.

2.Voting

Voting in sci-kit-learn (Sklearn) allows us to combine multiple machine-learning modules and use a majority vote or a weighted vote to make predictions. It's a way to ensemble different models for potentially better performance.

The Voting in Sklearn is an ensemble method that combines multiple individual classifiers or regressors to make predictions. It supports both hard and soft voting strategies. Now let's understand what ensemble learning is.

Voting Strategies:

Hard Voting – The class that receives the majority of votes is selected as the final prediction. It is commonly used in classification problems. In regression, it predicts the average of the individual predictions.

Soft Voting – Weighted average of predicted probabilities is used to make the final prediction. It is suitable when classifiers provide probability estimates. In other words, for each class, it sums the predicted probabilities and predicts the class with the highest sum.

Analysis of ensemble Technique

Ensemble Technique	Test Accuracy
Voting	97.09
Stacking	96.89

Implementation

Here we have provided classification Report of each ML Algorithm

1.GaussianNB

```
GaussianNB Test Accuracy: 0.9038666666666667
              precision    recall  f1-score   support

           0       0.97      0.93      0.95     13724
           1       0.45      0.65      0.54      1276

    accuracy                           0.90     15000
   macro avg       0.71      0.79      0.74     15000
weighted avg       0.92      0.90      0.91     15000
```

2.KNN (kneighborsClassifier)

```
KNeighborsClassifier Test Accuracy: 0.9622666666666667
              precision    recall  f1-score   support

           0       0.97      0.99      0.98     13724
           1       0.92      0.61      0.73      1276

    accuracy                           0.96     15000
   macro avg       0.94      0.80      0.86     15000
weighted avg       0.96      0.96      0.96     15000
```

3.Logistic Regression

```
LogisticRegression Test Accuracy: 0.9613333333333334
              precision    recall  f1-score   support

           0       0.97      0.99      0.98     13724
           1       0.88      0.63      0.74      1276

    accuracy                           0.96     15000
   macro avg       0.92      0.81      0.86     15000
weighted avg       0.96      0.96      0.96     15000
```

4.Decision Tree Classifier

```
DecisionTreeClassifier Test Accuracy: 0.9726666666666667
              precision    recall  f1-score   support

           0       0.97      1.00      0.99     13724
           1       1.00      0.68      0.81      1276

    accuracy                           0.97     15000
   macro avg       0.99      0.84      0.90     15000
weighted avg       0.97      0.97      0.97     15000
```

5.Random Forest Classifier

```
RandomForestClassifier Test Accuracy: 0.9726666666666667
              precision    recall  f1-score   support

           0       0.97      1.00      0.99     13724
           1       1.00      0.68      0.81      1276

    accuracy                           0.97     15000
   macro avg       0.99      0.84      0.90     15000
weighted avg       0.97      0.97      0.97     15000
```

6.SVC (Support Vector Machine)

```
SVC Test Accuracy: 0.9612666666666667
              precision    recall  f1-score   support

           0       0.96      1.00      0.98     13724
           1       0.92      0.60      0.72      1276

    accuracy                           0.96     15000
   macro avg       0.94      0.80      0.85     15000
weighted avg       0.96      0.96      0.96     15000
```

7.Voting Classifier

```
Ensemble Model Accuracy (Voting Classifier): 97.09%
              precision    recall  f1-score   support

           0       0.97      1.00      0.98     18292
           1       0.96      0.69      0.80      1708

    accuracy                           0.97     20000
   macro avg       0.97      0.84      0.89     20000
weighted avg       0.97      0.97      0.97     20000
```

8.Stacking Classifier

```
Stacking Model Accuracy: 96.89%
              precision    recall  f1-score   support

           0       0.97      0.99      0.98     18292
           1       0.92      0.69      0.79      1708

    accuracy                           0.97     20000
   macro avg       0.95      0.84      0.89     20000
weighted avg       0.97      0.97      0.97     20000
```

Additional Feature

1.Report Analysis

By using the **LIME**(Local Interpretable Model-Agnostic Explanation) library of python ,which is one type of explainable AI . It allows users to understand why a machine learning model made a particular decision. By adding model explainability into our project , we can make it more user friendly and credible. Which is more useful in healthcare, where decisions must be cleared.

Example

HbA1c_level > 6.20: 0.5553

140.00 < blood_glucose_level <= 159.00: -0.0690

bmi <= 23.65: -0.0405

heart_disease <= 0.00: -0.0334

hypertension <= 0.00: -0.0281

age <= 24.00: -0.0274

3.00 < smoking_history <= 4.00: 0.0150

0.00 < gender <= 1.00: 0.0042

In the above example we can see that the features name (HbA1c level, bmi, etc) represents the values entered by the user . lime generates average values by analyzing the data set and comparing the values with user entered values . and tells how much that value contributes to the prediction made by model.

2.personal Recommendation

Instead of just predicting whether a user has diabetes or not , we also provide personalized recommendations based on users input data.

Example

If the BMI is high, recommend a healthy weight-loss plan.

If the glucose levels are borderline, suggest dietary changes or exercise.

Conclusion

One of the important real-world medical problems is the detection of diabetes at its early stage. In this study, systematic efforts are made in designing a system which results in the prediction of diabetes.

In this research, we have conducted a comprehensive study on diabetes detection using ML techniques, aiming to underscore both the scientific value added by our work and the applicability of our findings in clinical practice. Our contributions encompass the outcome of an optimized preprocessing pipeline, addressing dataset imbalance, preventing overfitting, and demonstrating superior performance through extensive experimentation. We have rigorously evaluated various ML models on datasets Our results showcase notable improvements in accuracy, precision, recall, and F1-score metrics compared to existing methods.

In future, the designed system with the used machine learning classification algorithms can be used to predict or diagnose other diseases. The work can be extended and improved for the automation of diabetes analysis including some other machine learning algorithms.